VELTRAXX'26

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते

Chips to Startup Programme

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)
BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA

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TEAM NAME: MH Titans

PROBELM STATEMENT NO: PS16

PROBLEM STATEMENT TITLE: FPGA based Railway Interlocking

PS Category- Software/Hardware

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PROBLEM STATEMENT

REAL-WORLD ISSUE

- Multiple trains can simultaneously request access to routes.
- Physical trains take time to navigate through routes.
- Track switches require mechanical clearance delays.
- Track occupancy must be monitored before a route is released.

IMPORTANCE

- enter a fail-safe stateConflicting paths are a major collision hazard.
- Route locking must remain active until the track is confirmed clear.
- Emergency requests must preempt active routes safely before activation.
- Any timed-out condition must deny access and .

PRESENT CHALLENGES

- Basic FPGA demonstrations often use simple button-to-LED interactions.
- This logic lacks state memory and sequencing.
- It cannot maintain route locks after a button is released.
- It cannot safely handle emergency preemption, clearance delays, and watchdog faults.

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SLIDE 3 — HOW IT SOLVES THE PROBLEM

🔒 Route Locking

- **FSM** replaces simple combinational logic.
- Route remains **LOCKED** after button release.
- Route releases only after **Track Clear**.

🚨 Emergency Preemption

- Active routes are immediately forced to **RED**.
- **3-second all-RED clearance delay** is enforced.
- Emergency route turns **GREEN only after clearance.**

Watchdog Fail-Safe

- **10-second watchdog** detects stalled trains.
- Timeout → global **SYSTEM FAULT** with blinking RED.
- Fault remains active until **manual reset**.

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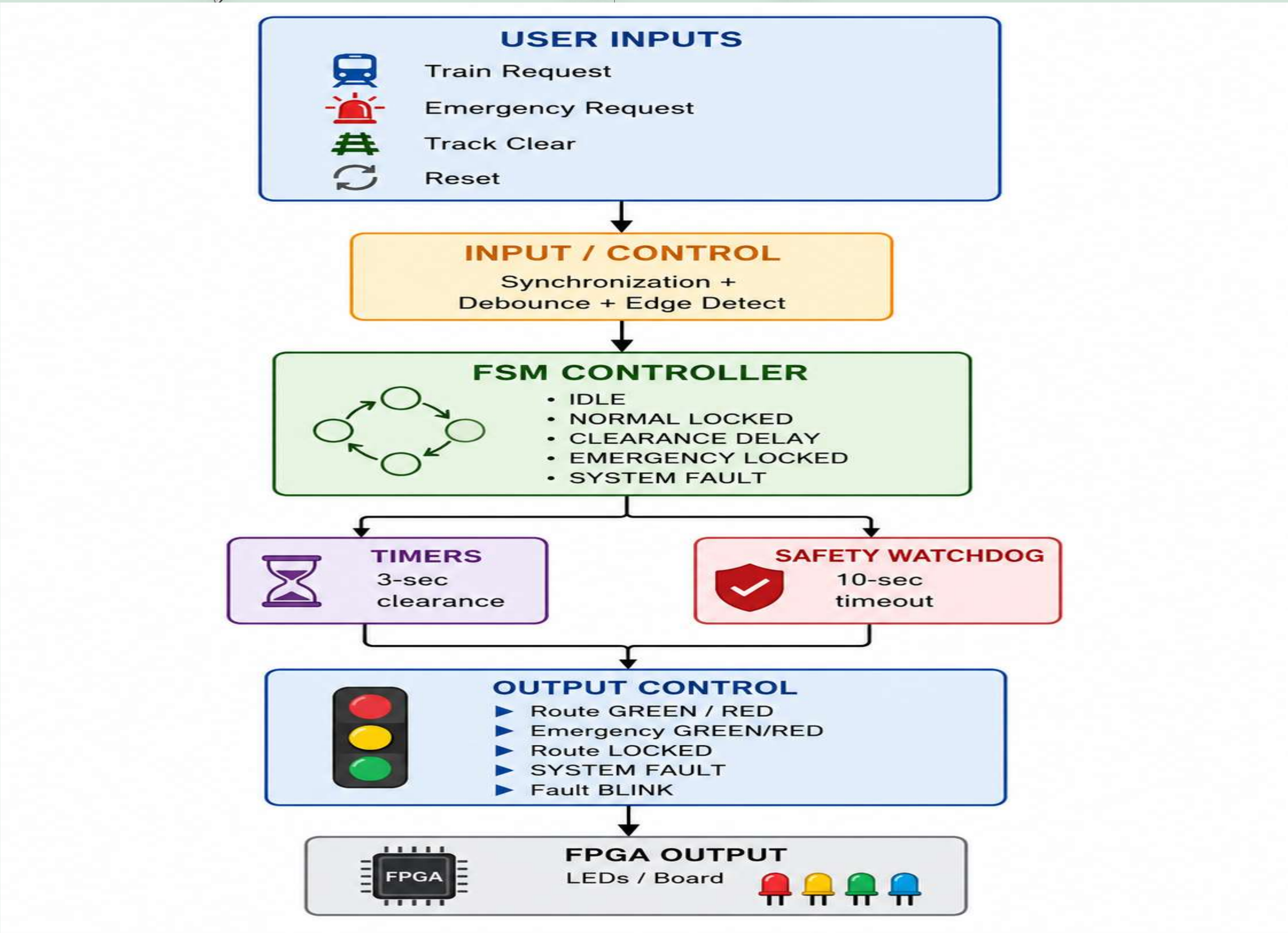
USER INPUTS
Train Request
Emergency Request
Track Clear
Reset
INPUT / CONTROL
Synchronization +
Debounce + Edge Detect
FSM CONTROLLER
• IDLE
• NORMAL LOCKED
• CLEARANCE DELAY
• EMERGENCY LOCKED
• SYSTEM FAULT
TIMERS
3-sec
clearance
SAFETY WATCHDOG
10-sec
timeout
OUTPUT CONTROL
► Route GREEN / RED
► Emergency GREEN/RED
► Route LOCKED
► SYSTEM FAULT
► Fault BLINK
FPGA OUTPUT
LEDs / Board
FPGA

PURPOSE OF BLOCKS

User Inputs

- Captures **Train Request, Emergency Request, Track Clear & Reset**.
- Provides the required control inputs to the system.

Input Control

- **Synchronizes, debounces and edge-detects** input signals.
- Prevents false or repeated requests.

FSM Controller

- Provides **state memory and sequential control**.
- Maintains route **LOCK** until Track Clear.

Clearance Timer

- Generates the **3-second all-RED delay**.
- Models track-switch clearance time.

Safety Watchdog

- Monitors active routes for **10 seconds**.
- Timeout → **SYSTEM FAULT.**

Output Control

- Converts FSM states into **GREEN/RED, LOCKED and FAULT** indications.
- LEDs represent railway signals for the hardware demo.

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OUTPUTS AND RESULTS

Normal route locking ✓
Locks on request, releases only on track clear

Emergency preemption ✓
All-red instantly, 3s delay, then emergency green

Watchdog timeout ✓
No track clear for 10s triggers fault

Fault indication ✓
System fault LED blinks at ~1Hz until reset

All 4 behaviors hardware verified

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RESEARCH AND REFERENCES

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- Fantechi, A., Gnesi, S., & Bacci, G. (2022). *Verification of Railway Interlocking Systems*.
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